THE EMBEDDING OF EVENT STRUCTURES INTO EVENT STRUCTURES WITH SYMMETRY IS 2-DENSE

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Editorial note. Source https://arxiv.org/abs/2609.17324v1. Copyright Serge Lechenne and Hugo Paquet, CC BY4.0. Complete native author 2-density construction retained, including its explicit omitted-verification statement. Existing corpus210587 counts only prop:Qonmor; this item is the independent main event-structure 2-density theorem (conference Theorem36), not a duplicate or a new count for its supporting context.

Allowed standard prerequisites (editorial).

- Locally small 2-categories, enriched hom functors, strict natural transformations and modifications. Native 2-density definition and publisher reference17 Kelly; event-structure symmetry conventions from reference32 and actual native definitions included here.

Definition. *An* event structure *is a tuple* $(E, \leq, \#)$ *where* $\leq$ *is a partial order on* E*, and* $\#$ *is an irreflexive and symmetric binary relation on* E*, satisfying the axioms below:*

- *For every* $e \in E$ *the set* $\{e' \in E \mid e' \leq e\}$ *is finite.*
- *For* $e, e', e'' \in E$*, if* $e \leq_E e'$ *and* $e \# e''$ *then* $e' \# e''$*.*

The execution paths of an event structure are called *configurations*:

Definition. *A finite subset* $X \subseteq E$ *is called* consistent *if no two events in* X *are in conflict. It is called a* configuration *if it is additionally down-closed. The set of consistent subsets of* E *is denoted* $\mathrm{Con}(E)$*, and the set of configurations is called* $\mathscr{C}(E)$*.*

Maps of event structures must faithfully preserve configurations:

Definition. *If* $(E, \leq_E, \#_E)$ *and* $(E', \leq_{E'}, \#_{E'})$ *are event structures, a function* $f : E \to E'$ *is a* map of event structures *if for every* $x \in \mathscr{C}(E)$*, the direct image* $fx := \{f(e) \mid e \in x\}$ *is in* $\mathscr{C}(E')$*, and moreover the restriction of* f *to* x *is injective.*

Event structures and maps of event structures form a category **Ev**. Frequently, we will regard **Ev** as a locally discrete 2-category, just like we did for **Petri** and **Occ**.

0.1. **The unfolding of a Petri net as an event structure with symmetry.** We first recall event structures with symmetry and the 2-category **EvSym**, originally introduced in [32].

0.1.1. *Event structures with symmetry.* Informally, symmetry on an event structure indicates which pairs of executions can be considered equivalent, via a bisimulation relation.

Definition. *For an event structure* E*, an isomorphism family is a set* $\mathbb{S}$ *of bijections* $\theta : x \cong y$ *between finite configurations of* $x, y \in \mathscr{C}(E)$*, such that:*

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- $\mathbb{S}$ *contains all identity bijections, and is stable under composition and inverse.*
- *For* $\theta : x \cong y \in \mathbb{S}$*, if* $x' \subseteq x$*, then the restriction of* θ *to* x' *is in* $\mathbb{S}$*.*
- *For* $\theta : x \cong y \in \mathbb{S}$*, if* $x \subseteq x'$*, then there exists an extension* $y \subseteq y'$ *and a bijection* $\theta' : x' \cong y' \in \mathbb{S}$ *such that* θ' *restricts to* θ*.*

The pair $(E, \mathbb{S})$ *is called an* event structure with symmetry*.*

We then consider symmetry-preserving maps:

Definition. *For event structures with symmetry* $(E, \mathbb{S})$ *and* $(E', \mathbb{S}')$*, a map* $f : E \to E'$ preserves symmetry *if for all* $\theta : x \cong y \in \mathbb{S}$*,* $f\theta : fx \cong fy \in \mathbb{S}'$*, where* $f\theta : f(e) \mapsto f(\theta(e))$*.*

A novelty in the presence of symmetry is the following equivalence relation on maps:

Definition. *For event structures with symmetry* $(E, \mathbb{S})$ *and* $(E', \mathbb{S}')$ *and symmetry-preserving maps* $f, g : E \to E'$*, say that* f *and* g *are* symmetric*, denoted* $f \sim g$*, if for every* $x \in \mathscr{C}(E)$ *the bijection* $fx \cong gx$ *defined by* $f(e) \mapsto g(e)$ *is in* $\mathbb{S}'$*.*

Together, event structures with symmetry, symmetry-preserving maps and symmetries of maps form a 2-category **EvSym** [32]. This is a degenerate 2-category in the sense that hom-categories are all *setoids* (sets with an equivalence relation). There is an embedding $J : \mathbf{Ev} \to \mathbf{EvSym}$. As we will see, the 2-categorical structure makes it possible to consider universal properties "up to symmetry".

0.2. **A 2-density result for event structures.** We prove a density result for the embedding of event structures into event structures with symmetry. This is a 2-categorical (i.e. **Cat**-enriched) form of density: each event structure with symmetry is a canonical 2-colimit of ordinary event structures. Another way of stating this is as follows:

Definition (e.g. [17]**.** *] For locally small 2-categories* $\mathbf{D}$ *and* $\mathbf{E}$*, a 2-functor* $J : \mathbf{D} \to \mathbf{E}$ *is* 2-dense *if the 2-functor* $\tilde{J} : \mathbf{E} \to [\mathbf{D}^{\mathrm{op}}, \mathbf{Cat}] : e \mapsto \mathbf{E}(J-, e)$ *is locally an isomorphism of categories, where* $[\mathbf{D}^{\mathrm{op}}, \mathbf{Cat}]$ *is the 2-category of 2-functors, strict natural transformations, and modifications.*

Theorem. *The embedding 2-functor* $J : \mathbf{Ev} \to \mathbf{EvSym}$ *is 2-dense.*

(We emphasize that 2-density is strictly weaker than density in the 1-categorical sense. In particular the underlying 1-functor $J : \mathbf{Ev} \to \mathbf{EvSym}$ is not dense.)

Proof. For event structures with symmetry A and B, we show that the functor $\tilde{J}_{A,B} : \mathbf{EvSym}(A, B) \to \mathbf{Cat}^{\mathbf{Ev}^{\mathrm{op}}}(\mathbf{EvSym}(J-, A), \mathbf{EvSym}(J-, B))$ is an isomorphism of categories by constructing an inverse. Write $A = (|A|, \mathbb{S}_A)$ and $B = (|B|, \mathbb{S}_B)$.

Observe that, for a natural transformation $\alpha : \mathbf{EvSym}(J-, A) \to \mathbf{EvSym}(J-, B)$, applying component $\alpha_{|A|}$ to the 'identity' function $\mathrm{id}_{|A|} : J|A| \to A$ gives a map of event structures with symmetry $\alpha_{|A|}(\mathrm{id}_{|A|}) : J|A| \to B$. We claim that $\alpha_{|A|}(\mathrm{id}_{|A|})$ preserves the symmetry in A. For any bijection $\theta : x \cong y$ in $\mathbb{S}_A$, we have $(\mathrm{id}_{|A|} \circ J\iota_y \circ J\theta) \sim (\mathrm{id}_{|A|} \circ J\iota_x)$, writing $\iota_x : x \to |A|$ and $\iota_y : y \to |A|$ for the inclusion maps (and recalling that $\theta \in \mathbf{Ev}(x, y)$). Since α is a natural transformation and each component preserves the symmetry relation on maps, $\alpha_{|A|}(\mathrm{id}_{|A|}) \circ J\iota_x = \alpha_x(\mathrm{id}_{|A|} \circ J\iota_x) \sim \alpha_x(\mathrm{id}_{|A|} \circ J\iota_y \circ J\theta) = \alpha_{|A|}(\mathrm{id}_{|A|}) \circ J\iota_y \circ J\theta$. In other words, $\alpha_{|A|}(\mathrm{id}_{|A|})\theta \in \mathbb{S}_B$, and we have proven the claim. Therefore, we take $\tilde{J}^{-1}_{A,B}(\alpha) = \alpha_{|A|}(\mathrm{id}_{|A|})$. For functoriality of $J^{-1}_{A,B}$, note that any modification $\alpha \to \beta$ must be unique and gives $\alpha_{|A|}(\mathrm{id}_{|A|}) \sim \beta_{|A|}(\mathrm{id}_{|A|})$. We omit the verification that $\tilde{J}^{-1}_{A,B}$ and $\tilde{J}_{A,B}$ are inverses. □

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